Chapter 4. Inference in Simple Regression

Sampling distributions, confidence intervals, prediction, inference on correlation, residuals

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Throughout, $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ with $\epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$ and fixed x_i with $S_{xx} > 0$.

4.1 Sampling distributions of the estimators

Motivation. The fitted slope would change with a new sample. To judge whether an observed slope reflects a real relationship we need its sampling distribution; under normal errors $\hat\beta_1$ is normal and the studentized slope has a t distribution.

Both estimators are linear in y:

$$\hat\beta_1 = \frac{S_{xy}}{S_{xx}} = \sum_i k_i y_i, \qquad k_i = \frac{x_i - \bar x}{S_{xx}}, \qquad \sum k_i = 0, \quad \sum k_i x_i = 1, \quad \sum k_i^2 = \frac{1}{S_{xx}},$$

(using $S_{xy} = \sum (x_i - \bar x) y_i$), and $\hat\beta_0 = \bar y - \hat\beta_1 \bar x$.

Theorem 4.1

(a) $\hat\beta_1 \sim N(\beta_1, \sigma^2/S_{xx})$.

(b) $\operatorname{Cov}(\bar y, \hat\beta_1) = 0$.

(c) $\hat\beta_0 \sim N\Big(\beta_0,\ \sigma^2\Big(\frac{1}{n} + \frac{\bar x^2}{S_{xx}}\Big)\Big)$ and $\operatorname{Cov}(\hat\beta_0, \hat\beta_1) = -\dfrac{\bar x \sigma^2}{S_{xx}}$.

Proof. (a) $\hat\beta_1$ is a linear combination of independent normals, with mean $\sum k_i(\beta_0 + \beta_1 x_i) = \beta_1$ and variance $\sigma^2 \sum k_i^2 = \sigma^2/S_{xx}$.

(b) $\operatorname{Cov}(\bar y, \hat\beta_1) = \sum_i \frac{1}{n} k_i \sigma^2 = 0$.

(c) $E\hat\beta_0 = (\beta_0 + \beta_1 \bar x) - \bar x \beta_1 = \beta_0$ and, by (b),

$$\operatorname{Var} \hat\beta_0 = \operatorname{Var} \bar y + \bar x^2 \operatorname{Var} \hat\beta_1 = \sigma^2\Big(\frac{1}{n} + \frac{\bar x^2}{S_{xx}}\Big), \qquad \operatorname{Cov}(\hat\beta_0, \hat\beta_1) = -\bar x \operatorname{Var} \hat\beta_1.$$

■

Theorem 4.2

Let $SSE = \sum e_i^2$ and $MSE = SSE/(n-2)$. Then $SSE/\sigma^2 \sim \chi^2(n-2)$, independent of $(\hat\beta_0, \hat\beta_1)$. In particular $E(MSE) = \sigma^2$.

This is the case $p = 1$ of Theorem 6.4; a direct proof uses an orthogonal transformation whose first two rows span $\mathbf{1}$ and $\mathbf{x} - \bar x \mathbf{1}$, exactly as for the normal sample variance. Consequently

$$\frac{\hat\beta_1 - \beta_1}{\sqrt{MSE/S_{xx}}} \sim t(n-2), \qquad \frac{\hat\beta_0 - \beta_0}{\sqrt{MSE(1/n + \bar x^2/S_{xx})}} \sim t(n-2).$$

Corollary 4.3 (Intervals and tests for the coefficients)

$100(1-\alpha)\%$ confidence intervals are

$$\beta_1 :\ \hat\beta_1 \pm t_{\alpha/2}(n-2)\sqrt{\frac{MSE}{S_{xx}}}, \qquad \beta_0 :\ \hat\beta_0 \pm t_{\alpha/2}(n-2)\sqrt{MSE\Big(\frac{1}{n} + \frac{\bar x^2}{S_{xx}}\Big)}.$$

The test of $H_0 : \beta_1 = 0$ rejects when $|t_0| > t_{\alpha/2}(n-2)$, $t_0 = \hat\beta_1/\sqrt{MSE/S_{xx}}$. If σ^2 is known, replace MSE by σ^2 and t by z.

Theorem 4.4 (ANOVA F test)

$E(SSR) = \sigma^2 + \beta_1^2 S_{xx}$ and $E(SSE) = (n-2)\sigma^2$. The statistic $F_0 = MSR/MSE$, $MSR = SSR/1$, satisfies $F_0 = t_0^2$, so the F test with $F_\alpha(1, n-2) = t_{\alpha/2}(n-2)^2$ is the two-sided t test of $\beta_1 = 0$.

Proof. $E(SSR) = S_{xx}E\hat\beta_1^2 = S_{xx}(\sigma^2/S_{xx} + \beta_1^2)$. Also $SSR/MSE = \hat\beta_1^2 S_{xx}/MSE = t_0^2$. ■

So $E(MSR)/E(MSE) = 1 + \beta_1^2 S_{xx}/\sigma^2$, which is 1 exactly when $\beta_1 = 0$; large F_0 is evidence against H_0.

4.2 Estimating the mean response and predicting

Motivation. Estimating the average response at x_0 and predicting a single new response there are different problems: the prediction interval must also account for the new observation's own error, so it is always wider.

At a point x, the mean response $\mu_{y\cdot x} = \beta_0 + \beta_1 x$ is estimated by $\hat y = \hat\beta_0 + \hat\beta_1 x = \bar y + \hat\beta_1(x - \bar x)$.

Theorem 4.5

(a) $\hat y \sim N\Big(\mu_{y\cdot x},\ \sigma^2\Big(\dfrac{1}{n} + \dfrac{(x-\bar x)^2}{S_{xx}}\Big)\Big)$, and a $100(1-\alpha)\%$ confidence interval for $\mu_{y\cdot x}$ is

$$\hat y \pm t_{\alpha/2}(n-2)\sqrt{MSE\Big(\frac{1}{n} + \frac{(x-\bar x)^2}{S_{xx}}\Big)}.$$

(b) For a new observation $y_s = \beta_0 + \beta_1 x + \epsilon_s$, independent of the sample, $y_s - \hat y \sim N\big(0, \sigma^2(1 + \frac{1}{n} + \frac{(x-\bar x)^2}{S_{xx}})\big)$, and a $100(1-\alpha)\%$ prediction interval is

$$\hat y \pm t_{\alpha/2}(n-2)\sqrt{MSE\Big(1 + \frac{1}{n} + \frac{(x-\bar x)^2}{S_{xx}}\Big)}.$$

Proof. (a) By Theorem 4.1(b), $\operatorname{Var}\hat y = \operatorname{Var}\bar y + (x-\bar x)^2 \operatorname{Var}\hat\beta_1$. Studentize with MSE, which is independent of $\hat y$.

(b) y_s is independent of $\hat y$, so the variances add. ■

Both intervals are narrowest at $x = \bar x$ and widen quadratically away from it; extrapolation is costly. The prediction interval does not shrink to zero as $n \to \infty$, because of the irreducible variance σ^2 of a new observation.

Example 4.1

For the advertising data of Example 3.1 ($n = 6$, $\bar x = 5.667$, $S_{xx} = 45.33$, $MSE = 1.92/4 = 0.48$, $\hat y = 2.37 + 1.346x$), at $x = 7$: $\hat y = 11.79$, $\frac{1}{6} + \frac{(1.333)^2}{45.33} = 0.206$, and $t_{0.025}(4) = 2.776$. The 95% confidence interval for the mean is $11.79 \pm 2.776\sqrt{0.48 \cdot 0.206} = 11.79 \pm 0.87$, and the prediction interval is $11.79 \pm 2.776\sqrt{0.48 \cdot 1.206} = 11.79 \pm 2.11$.

4.3 Inference on the correlation

When (X, Y) is bivariate normal with correlation ρ, the conditional mean is linear with slope $\beta_1 = \rho\sigma_Y/\sigma_X$, so $\rho = 0 \iff \beta_1 = 0$.

Theorem 4.6

(a) The t statistic for $\beta_1 = 0$ equals $t_0 = \dfrac{r\sqrt{n-2}}{\sqrt{1-r^2}}$, which is $t(n-2)$ under $\rho = 0$.

(b) (Fisher) $Z = \frac{1}{2}\log\frac{1+r}{1-r}$ is approximately $N\big(\frac{1}{2}\log\frac{1+\rho}{1-\rho}, \frac{1}{n-3}\big)$ for moderate n (say $n \geq 25$).

Proof. (a) $SSE = SST(1-r^2)$ and $\hat\beta_1^2 S_{xx} = SSR = r^2 SST$, so

$$t_0^2 = \frac{\hat\beta_1^2 S_{xx}}{SSE/(n-2)} = \frac{(n-2)r^2}{1-r^2}.$$

The null distribution is conditionally $t(n-2)$ given the x_i, hence unconditionally as well.

(b) is the variance-stabilizing transformation of the asymptotic law $\sqrt{n}(r-\rho) \to N(0, (1-\rho^2)^2)$; the correction $n-3$ improves the approximation. ■

Part (b) gives tests of $H_0 : \rho = \rho_0 \neq 0$ and confidence intervals: $\tanh\big(Z \pm z_{\alpha/2}/\sqrt{n-3}\big)$ covers ρ with probability about $1-\alpha$.

4.4 Residuals

Motivation. All inference rests on model assumptions. Residuals are our window on the unobservable errors, and plotting them is how we check linearity, constant variance and normality.

Residuals are our window on the errors. In vector form $\mathbf{e} = (I - H)\mathbf{y}$ with H the hat matrix (Chapter 6), and:

Theorem 4.7

$Ee_i = 0$ and

$$\operatorname{Var}(e_i) = \sigma^2\Big(1 - \frac{1}{n} - \frac{(x_i-\bar x)^2}{S_{xx}}\Big), \qquad \operatorname{Cov}(e_i, e_j) = -\sigma^2\Big(\frac{1}{n} + \frac{(x_i-\bar x)(x_j-\bar x)}{S_{xx}}\Big) \ (i \neq j).$$

Proof. $e_i = y_i - \bar y - \hat\beta_1(x_i - \bar x)$. Using $\operatorname{Var}\bar y = \sigma^2/n$, $\operatorname{Var}\hat\beta_1 = \sigma^2/S_{xx}$, $\operatorname{Cov}(y_i, \bar y) = \sigma^2/n$, $\operatorname{Cov}(y_i, \hat\beta_1) = \sigma^2 k_i$ and $\operatorname{Cov}(\bar y, \hat\beta_1) = 0$,

$$\operatorname{Var} e_i = \sigma^2\Big(1 + \frac{1}{n} + \frac{(x_i-\bar x)^2}{S_{xx}} - \frac{2}{n} - 2\frac{(x_i-\bar x)^2}{S_{xx}}\Big),$$

which simplifies to the claim; the covariance is computed in the same way. ■

Thus residuals are neither independent nor equally variable; points far from $\bar x$ have smaller residual variance (Chapter 10). Since $\sum e_i = 0$ and $\sum x_i e_i = 0$, residual plots against x_i or $\hat y_i$ have no linear trend by construction; what we look for is *non-random* structure:

- a curved band suggests a missing quadratic term or a transformation;
- a funnel suggests nonconstant variance (weighted least squares or a variance-stabilizing transformation of y);
- a trend in time order suggests correlated errors (Chapter 12).

4.5 Selected exercises

Motivation. The problems below add what the sampling theory above leaves implicit: when the two coefficient estimators are independent, why a one-sided t test has the right size over a composite null, how $E(SSE)$ follows from the residual variances alone, and how the design controls $\operatorname{Var}\hat{\beta}_1$.

Exercise 4.1 (Independence of the coefficient estimators)

Show that $\bar{y}$ and $\hat{\beta}_1$ are always independent, and that $\hat{\beta}_0$ and $\hat{\beta}_1$ are independent if and only if $\bar{x} = 0$.

Solution.

Step 1. (Joint normality.) Each of $\bar{y}$, $\hat{\beta}_0$, $\hat{\beta}_1$ is a fixed linear combination of $\mathbf{y} = (y_1, \dots, y_n)'$, so $(\hat{\beta}_0, \hat{\beta}_1, \bar{y})' = A\mathbf{y}$ for a fixed $3 \times n$ matrix A. Since $\mathbf{y} \sim N_n(X\beta, \sigma^2 I)$, $A\mathbf{y}$ is multivariate normal (Section 5.4).

Step 2. (Uncorrelated implies independent.) For a jointly normal pair (U, W) with $\operatorname{Cov}(U, W) = 0$, the mgf is $\exp\{t_1 EU + t_2 EW + \frac{1}{2}(t_1^2 \operatorname{Var} U + t_2^2 \operatorname{Var} W)\}$, the product of the two marginal mgfs; hence U and W are independent. Conversely, independence implies zero covariance.

Step 3. By Theorem 4.1(b), $\operatorname{Cov}(\bar{y}, \hat{\beta}_1) = 0$, so $\bar{y}$ and $\hat{\beta}_1$ are independent for every design.

Step 4. By Theorem 4.1(c), $\operatorname{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\bar{x}\sigma^2/S_{xx}$, which is zero if and only if $\bar{x} = 0$. By Step 2, $\hat{\beta}_0$ and $\hat{\beta}_1$ are independent if and only if $\bar{x} = 0$. (If $\bar{x} = 0$, then $\hat{\beta}_0 = \bar{y}$, in line with Step 3.) ■

Exercise 4.2 (A one-sided test with a composite null)

Let $0 < \alpha < 1$ and test $H_0 : \beta_1 \le 0$ against $H_1 : \beta_1 > 0$. Show that the test rejecting H_0 when

$$t_0 = \frac{\hat{\beta}_1}{\sqrt{MSE/S_{xx}}} > t_\alpha(n-2)$$

satisfies

$$\sup_{\beta_0 \in \mathbb{R},\ \beta_1 \le 0,\ \sigma^2 > 0} P_{\beta_0, \beta_1, \sigma^2}(\text{the test rejects } H_0) = \alpha,$$

and that its rejection probability exceeds α whenever $\beta_1 > 0$.

Solution.

Step 1. For every parameter value, $T = (\hat{\beta}_1 - \beta_1)/\sqrt{MSE/S_{xx}} \sim t(n-2)$ (Section 4.1). Since $MSE > 0$ with probability one,

$$t_0 = T + \frac{\beta_1}{\sqrt{MSE/S_{xx}}}.$$

Step 2. If $\beta_1 \le 0$, the last term is ≤ 0, so $t_0 \le T$ and $\{t_0 > t_\alpha(n-2)\} \subseteq \{T > t_\alpha(n-2)\}$. Hence the rejection probability is at most $P(T > t_\alpha(n-2)) = \alpha$.

Step 3. If $\beta_1 = 0$, then $t_0 = T$ and the rejection probability is exactly α, for every β_0 and σ^2. With Step 2, the supremum over the null set $\{\beta_1 \le 0\}$ equals α and is attained on the boundary $\beta_1 = 0$.

Step 4. (Power.) $\hat{\beta}_1 \sim N(\beta_1, \sigma^2/S_{xx})$ is independent of MSE (Theorem 4.2). Write $c = t_\alpha(n-2)$ and $\tau = \sigma/\sqrt{S_{xx}}$. Conditioning on MSE,

$$P(t_0 > c) = E\Big[P\big(\hat{\beta}_1 > c\sqrt{MSE/S_{xx}} \mid MSE\big)\Big] = E\Big[1 - \Phi\Big(\frac{c\sqrt{MSE/S_{xx}} - \beta_1}{\tau}\Big)\Big].$$

The integrand is strictly increasing in β_1, so the rejection probability is a strictly increasing function of β_1. It equals α at $\beta_1 = 0$ by Step 3, hence exceeds α for every $\beta_1 > 0$: the test is unbiased. (This also reproves Step 2.) ■

Exercise 4.3 (Expected residual sum of squares without chi-square theory)

(a) Using only Theorem 4.7, show that $E(SSE) = (n-2)\sigma^2$.
(b) Show that $\operatorname{Var}\hat{y}_i + \operatorname{Var} e_i = \sigma^2$ for each i, where $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$, and that $\sum_i \operatorname{Var}\hat{y}_i = 2\sigma^2$.

Solution [Supplement].

(a) Since $Ee_i = 0$, $Ee_i^2 = \operatorname{Var} e_i$. Summing the formula of Theorem 4.7 and using $\sum(x_i - \bar{x})^2 = S_{xx}$,

$$E(SSE) = \sum_i Ee_i^2 = \sigma^2\Big(n - 1 - \frac{S_{xx}}{S_{xx}}\Big) = (n-2)\sigma^2.$$

No normality is needed: only $E\epsilon = \mathbf{0}$ and $\operatorname{Var}\epsilon = \sigma^2 I$ were used.

(b) By Theorem 4.5(a) at $x = x_i$, $\operatorname{Var}\hat{y}_i = \sigma^2\{1/n + (x_i - \bar{x})^2/S_{xx}\}$; adding $\operatorname{Var} e_i$ from Theorem 4.7 gives σ^2. Summing over i,

$$\sum_i \operatorname{Var}\hat{y}_i = \sigma^2\Big(1 + \frac{S_{xx}}{S_{xx}}\Big) = 2\sigma^2.$$

So the total variance $n\sigma^2$ of the y_i splits into $2\sigma^2$ in the fitted values (two parameters) and $(n-2)\sigma^2$ in the residuals. ■

Exercise 4.4 (The best design for the slope)

Let n be even, and suppose the experimenter may choose each x_i in a fixed interval $[a, b]$. Show that

$$\operatorname{Var}\hat{\beta}_1 \geq \frac{4\sigma^2}{n(b-a)^2},$$

with equality if and only if half of the x_i equal a and the other half equal b.

Solution [Supplement].

Step 1. For any constant c, $\sum(x_i - c)^2 = S_{xx} + n(\bar{x} - c)^2 \geq S_{xx}$ (expand $x_i - c = (x_i - \bar{x}) + (\bar{x} - c)$; the cross term vanishes). Take $c = m = (a+b)/2$.

Step 2. Since $x_i \in [a, b]$, $|x_i - m| \leq (b-a)/2$. Hence

$$S_{xx} \leq \sum(x_i - m)^2 \leq \frac{n(b-a)^2}{4}, \qquad \operatorname{Var}\hat{\beta}_1 = \frac{\sigma^2}{S_{xx}} \geq \frac{4\sigma^2}{n(b-a)^2}.$$

Step 3. Equality holds in both inequalities of Step 2 if and only if $\bar{x} = m$ and $|x_i - m| = (b-a)/2$ for all i, i.e. every $x_i \in \{a, b\}$ and $\bar{x} = m$. If n_a points are at a and $n - n_a$ at b, then $\bar{x} = m$ means $n_a a + (n - n_a)b = n(a+b)/2$, i.e. $n_a = n/2$. Conversely the half-and-half design gives $S_{xx} = n(b-a)^2/4$ (Exercise 3.2).

Such a design cannot reveal curvature, since $E(y)$ is observed at two points only; in practice some points are placed in the interior to check linearity. ■